

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

January 2023

20EC504 – VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

Marks BT* CO* PO*

1. a) Design CMOS circuit for the logic gate shown in Figure Q.1a_1. For the circuit shown in Figure Q.1a_2, complete the following table and also determine the expression for the output, Y.

A	B	C	D	Pull up network	Pull down network	Y
0	0	0	0	ON	OFF	1
1	0	0	1			
1	1	1	0			

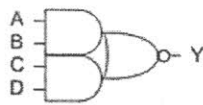


Figure Q.1a_1

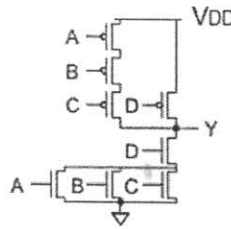


Figure Q.1a_2

- b) With relevant cross sectional view of MOSFET and waveforms, explain channel length modulation and velocity saturation non-idealities in MOSFETs.

10 L*3 1 1

10 L2 1 1

2. a) Determine the expression for F1 and F2 for the circuit shown in Figure Q.2a.

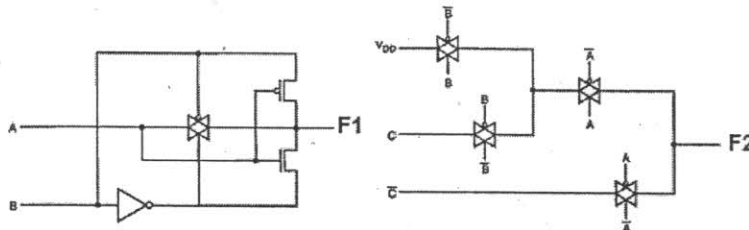


Figure Q.2a

- b) Derive the expression for the CMOS inverter threshold voltage, V_{inv} . Determine the device sizes for $V_{inv} = 0.5 V_{DD}$. Also define noise margin for LOW and noise margin for HIGH.

10 L3 1 1

10 L2 1 1

Unit – II

3. a) With neat sketches, write the lambda based rules for poly, diffusion, metals and contacts. Also design the stick diagram and layout for 3-input NAND gate.
b) Draw neat sketches of cross sectional view for the fabrication of a CMOS inverter using n-well process.

10 L3 2 1

10 L2 2 1

P.T.O.

4. a) Write the salient steps in Euler's graph method to design an area efficient layout. Using Euler's graph method, design the layout for a compound gate defined by $y_4 = (P + Q + R)S$.
- b) Write the constant field and constant voltage scaling models. Derive the effects of these scaling models on MOSFET power density.

10	L3	2	1
10	L2	2	1

Unit – III

5. a) Using Elmore delay model, compute t_{pdr} and t_{pdr} in a 3 input NAND gate.
- b) Calculate the path delays for the two cascades shown in Figure Q.5b, assuming electrical effort, $H=1$.

10	L2	3	1
----	----	---	---

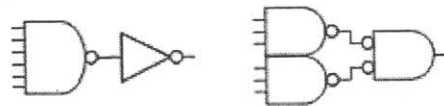


Figure Q.5b.

10	L3	3	1
----	----	---	---

6. a) Using Logical effort technique, show that the minimum path delay in a N-stage cascade with path effort, F and path parasitic delay, P is given by, $D=NF^{1/N} + P$.
- b) Explain pseudo-NMOS logic and CVS logic. Design Pseudo-NMOS NAND circuit and AND/NAND CVSL circuit.

10	L2	3	1
----	----	---	---

10	L3	3	1
----	----	---	---

Unit – IV

7. a) How dynamic logic circuits differ from static logic circuits? Also explain the shift operation in a ratioless dynamic shift register.
- b) Draw the dynamic CMOS circuit for 3 input NAND gate. Also explain the problem in cascading dynamic gates and suggest the solution.
8. a) Draw the logic diagram for the ring oscillator to generate on-chip clock. Also explain how global clocks are distributed in a chip.
- b) Explain the operation of D-flip flop with i) Asynchronous RESET ii) Asynchronous SET and RESET.

10	L2	4	1
----	----	---	---

10	L2	4	1
----	----	---	---

10	L2	4	1
----	----	---	---

10	L2	4	1
----	----	---	---

Unit – V

9. a) With reference to testing, explain observability, controllability, repeatability and survivability.
- b) Explain the READ and WRITE operation in a 1-T DRAM cell.
10. a) With relevant waveforms, explain READ and WRITE operations in a CMOS SRAM cell.
- b) What is design for testability? Also explain BIST procedure along with the description of LFSR as a pseudo-random pattern generator and ORA.

10	L2	5	1
----	----	---	---

10	L2	5	1
----	----	---	---

10	L2	5	1
----	----	---	---

10	L2	5	1
----	----	---	---